I Induced Norms

$\|x\|_2 = [\langle x, x \rangle]^{1/2}$

A) Examples

Let $R \succeq 0$ (positive definite matrix), and symmetric

(i.e. $x^T R x > 0 \quad \forall \; x \neq 0$)

Define $\langle , \rangle : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$

by $\langle x, y \rangle = \underline{y}^T R \underline{x}$

Use this inner product to define a norm

$\|x\|_2^2 = \langle x, x \rangle = x^T R x$

Now look at the gaussian

$(y - Ax)^T Q^{-1} (y - Ax) = \langle y - Ax, y - Ax \rangle$ where $R = Q^{-1}$

which can be thought of as $\underline{\|y - Ax\|_2^2}$

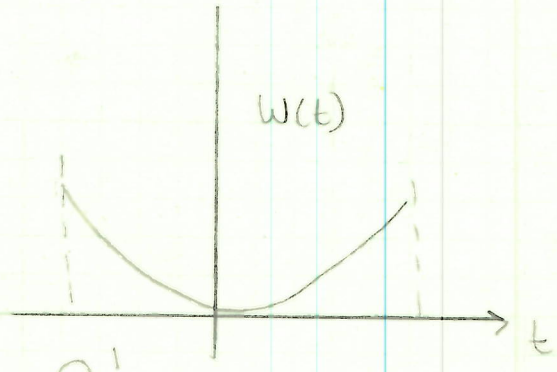
$L_2[-1, 1]$

$$\langle x(t), y(t) \rangle = \int_{-1}^{1} x(t) y(t) \, dt$$

Let $W(t)$ be a function

$W(t) \geq 0 \quad \forall \; t \in [-1, 1]$

except at possibly a countable number of points.



$$\langle x(t), y(t) \rangle = \int_{-1}^{1} w(t)\, x(t)\, y(t)\, d(t)$$

weighting factor

$$\| x(t) \|_2^2 = \langle x(t), x(t) \rangle$$

II Cauchy-Schwartz Inequality

$$\left[\int_a^b x(t) y(t)\, dt \right]^2 \leq \int_a^b x(t)^2\, dt + \int_a^b y(t)^2\, dt$$

More generally

$$|\langle x, y \rangle| \leq \|x\| \, \|y\|$$

where $\|\cdot\|$ is induced from $\langle \cdot, \cdot \rangle$ with equality iff $y = \alpha x$ for some scalar α.

In terms of sums

$$\left| \sum_{i=1}^{n} x_i y_i \right| \leq \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2} + \left(\sum_{i=1}^{n} |y_i|^2 \right)^2$$

Proof:

Consider $\| x - \alpha y \|^2$.

We know $\| x - \alpha y \|^2 \geq 0$

$$\|x - \alpha y\|^2 = \langle x - \alpha y, x - \alpha y \rangle$$
$$= \langle x, x \rangle - \langle x, \alpha y \rangle - \langle \alpha y, x \rangle + \langle \alpha y, \alpha y \rangle$$
$$= \|x\|^2 - \alpha^* \langle x, y \rangle - \alpha \langle y, x \rangle + |\alpha|^2 \langle y, y \rangle$$
$$= \|x\|^2 - 2\mathrm{Re}[\alpha^* \langle x, y \rangle] + |\alpha|^2 \|y\|^2$$

Complete the square in terms of α.

$$= \|y\|^2 \left(\alpha - \frac{\langle x, y \rangle}{\|y\|^2}\right)\left(\alpha^* - \frac{\langle y, x \rangle}{\|y\|^2}\right) + \|x\| - \frac{|\langle x, y \rangle|^2}{\|y\|^2}$$

Let $\alpha = \dfrac{\langle x, y \rangle}{\|y\|^2}$

True because this a norm

therefore, $\|x\|^2 = \dfrac{|\langle x, y \rangle|^2}{\|y\|^2} \geq 0$

$$\|x\|^2 \|y\|^2 \geq |\langle x, y \rangle|^2$$

Example

Now let $y = \alpha x$

$$|\langle x, y \rangle| = |\langle x, \alpha x \rangle| = |\alpha| \langle x, y \rangle = |\alpha| \|x\|^2$$

$$\|x\|^2 \|y\|^2 = \|x\|^2 \|\alpha x\|^2 = \|x\|^2 |\alpha|^2 \|x\|^2$$

taking the square root shows us that they are the same.

A) Think about

$$-1 < \frac{\langle x, y \rangle}{\|x\| \|y\|} < 1$$ ~ look like $\cos \theta$

Define an angle θ s.t. $\cos\theta = \dfrac{\langle x, y \rangle}{\|x\| \|y\|}$

Def:

If $x = \alpha y$, x and y are colinear

Def: If $\langle x, y \rangle = 0$, x and y are orthogonal, $x \perp y$

Def: If $V_1, V_2, \ldots, V_m$ and $\langle V_i, V_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$

then the set is orthonormal.

B) Example

$L_2[-1\ 1]$ $\quad x_1(t) = 1 + t^2$

$x_2(t) = t^2 - 2t$

$$\cos\theta = \frac{\langle x_1(t), x_2(t) \rangle}{\|x_1(t)\| \|x_2(t)\|} = \frac{\int_{-1}^{1} x_1(t) x_2(t)\, dt}{\left[\int_{-1}^{1} x_1(t)^2 dt\right]^{1/2} \left[\int_{-1}^{1} x_2(t)^2 dt\right]^{1/2}}$$

$L_2[-1\ 1]$ $\quad \langle x(t), y(t) \rangle = \int_{-1}^{1} w(t) x(t) y(t)\, dt$

$$W(t) = \frac{1}{\sqrt{1 - t^2}}$$

Let $T_n(t) = \cos(n \cos^{-1}(t))$ $\quad t \in [-1, 1]$ $\quad n = 0, 1, 2, \ldots$

$n = 0$ $\quad T_0(t) = \cos(0) = 1$

$n = 1$ $\quad T_1(t) = \cos(\cos^{-1}(t)) = t$

$n = 2$ $\quad T_2(t) = \cos(2\cos^{-1}(t)) = 2t^2 - 1$

$n = 3$ $\quad T_3(t) = \cos(3\cos^{-1}(t)) = 4t^3 - 3t$

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